### 8.1.2. Sources of Green House Gases Emission (Natural and Anthropogenic)

**$CO_2$ (Carbon dioxide)**

- Coal based power plants, by the burning of fossil fuels for electricity generation.
- Combustion of fuels in the engines of automobiles, commercial vehicles and air planes contribute the most of global warming.
- Agricultural practices like stubble burning result in emission of $CO_2$.
- Natural from organic matter, volcanoes, warm oceans and sediments.

**Methane**

Methane is 20 times as effective as $CO_2$ at trapping heat in the atomosphere. Its sources are attributed paddy cultivation, cattle rearing, bacteria in water bodies, fossil fuel production, ocean, non-wetland soils and forest / wild fires.

**$N_2O$ (Nitrous oxide)**

It is naturally produced in Oceans from biological sources of soil and water due to microbial actions and rainforests. Man-made sources include nylon and nitric acid production, use of fertilizers in agriculture, manures cars with catalytic converter and burning of organic matter.

**Global Warming Effects on Plants**

- Low agricultural productivity in tropics
- Frequent heat waves (Weeds, pests, fungi need warmer temperature)
- Increase of vectors and epidemics
- Strong storms and intense flood damage
- Water crisis and decreased irrigation
- Change in flowering seasons and pollinators
- Change in Species distributional ranges
- Species extinction

### 8.1.3 Strategies to deal with Global Warming

- Increasing the vegetation cover, grow more trees
- Reducing the use of fossil fuels and green house gases
- Developing alternate renewable sources of energy
- Minimising uses of nitrogeneous fertilizers, and aerosols.

### 8.1.4. Ozone depletion

Ozone layer is a region of Earth's stratosphere that absorbs most of the Sun's ultra violet radiation. The ozone layer is also called as the **ozone shield** and it acts as a protective shield, cutting the ultra-violet radiation emitted by the sun.

Just above the atmosphere there are two layers namely troposphere (the lower layer) and stratosphere (the upper layer). The ozone layer of the troposphere is called **bad ozone** and the ozone layer of stratosphere is known as **good ozone** because this layer acts as a shield for absorbing the UV radiations coming from the sun which is harmful for living organisms

Ozone is a colourless gas, reacts readily with air pollutants and cause rubber to crack, hurt plant life, damages lung tissues. But ozone absorbs harmful ultra violet β (uv-β) and UV – α radiation from sunlight.

What is Dobson Unit? DU is the unit of measurement for total ozone. One DU (0.001 atm. cm) is the number of molecules of ozone that would be required to create a layer of pure ozone 0.01 millimetre thick at a temperature of 0° C and a pressure of 1 atmosphere (atm = the air pressure at the surface of earth). Total ozone layer over the earth surface is 0.3 centrimetres (3 mm) thick and is written as 300 DU.

The false colour view of total ozone - The purple and blue colours are where there is the least ozone, and the yellows and reds are where there is more ozone.

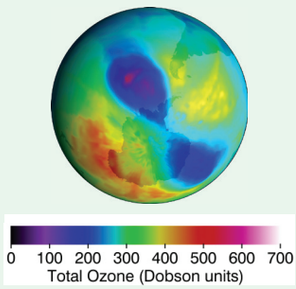



**Figure 8.3:** The false colour view of total ozone



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